Butter Churner



Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Butter Churner

What it is used for

A butter churner agitates cream to separate butterfat into butter granules and buttermilk. Use clean equipment, suitable cream conditions, and controlled churning to obtain consistent product quality while preventing contamination and mechanical injury.

Main parts

- Churning vessel or drum
- Lid and locking mechanism
- Agitator, paddles, or rotating churn assembly
- Drive motor, gearbox, and speed control
- Drain or buttermilk outlet
- Control panel, guards, and emergency stop

Before you start

1. Confirm the batch size, cream quality, and required processing conditions from the dairy procedure.
2. Inspect the vessel, lid, seals, agitator, outlet valve, drive system, and guards for damage or residue.
3. Ensure all product-contact surfaces have been cleaned and sanitised using approved food-safe procedures.
4. Check that the lid interlock, emergency stop, and speed controls work correctly.
5. Verify that the cream is within the specified temperature and quality range for churning.
6. Prepare clean collection containers for butter granules and buttermilk, and wear appropriate food-handling PPE.

How to use

1. Switch off and isolate power before opening the vessel, inspecting the agitator, or cleaning the machine.
2. Confirm that the drain valve is closed and that the vessel is clean and ready for the batch.
3. Measure and load the required quantity of cream without exceeding the manufacturer's capacity.
4. Secure the lid and confirm that the safety interlock is engaged where fitted.
5. Select the approved speed or operating setting and start the churner using the normal controls.
6. Observe the cream during churning and monitor the motor, product movement, temperature, and sound.

7. Continue churning until butter granules form and the separation stage meets the dairy procedure's criteria.
8. Stop the machine before opening the lid or draining buttermilk; wait for all moving parts to stop.
9. Drain the buttermilk through the designated outlet into a clean container without contaminating the butter.
10. Collect the butter granules and carry out washing, working, salting, or packaging only as specified by the product procedure.
11. Inspect the finished butter for expected texture and appearance and record yield and batch details.
12. After unloading, isolate power and clean, rinse, and sanitise the vessel, agitator, lid, and outlet according to the sanitation schedule.

⚠ Safety

- Only trained personnel should operate or maintain the churner.
- Never open the lid or reach into the vessel while the agitator is moving.
- Do not bypass lid interlocks, guards, or emergency stop devices.
- Keep hands, loose clothing, and tools away from rotating shafts and drive components.
- Use approved cleaning chemicals at the specified concentration and prevent residues from contacting food.
- Disconnect and isolate power before maintenance, jam clearing, or dismantling the agitator.

Care & storage

- Clean and sanitise the vessel, lid, agitator, and outlet after each batch.
- Inspect seals, bearings, drive belts, fasteners, and interlocks regularly.
- Check speed controls and motor operation according to the maintenance schedule.
- Keep the exterior and surrounding floor clean and dry to reduce slip hazards.
- Arrange servicing if the churner vibrates excessively, leaks, overheats, or produces unusual noise.

Common mistakes to avoid

- Loading too much cream or using cream outside the specified processing conditions.
- Churning at an incorrect speed or stopping before the required separation stage.
- Opening the lid or draining the vessel while the agitator is moving.
- Failing to clean and sanitise the machine between batches.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.